



Advanced Time Series Analysis and Forecasting

Chapter 1: Forecast evaluation, scoring rules and combination



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material (1/2)

- ▣ **AI**: Artificial Intelligence
- ▣ **ALFRED**: Archival Federal Reserve Economic Data
- ▣ **AO**: Atkeson–Ohanian (naive inflation forecast)
- ▣ **AR**: AutoRegressive (model); in event studies: Abnormal Return
- ▣ **ARX**: AutoRegressive model with eXogenous variables
- ▣ **BNR**: Banca Națională a României (the National Bank of Romania)
- ▣ **CPI**: Consumer Price Index
- ▣ **CRPS**: Continuous Ranked Probability Score
- ▣ **CW**: Clark–West (test for nested models)
- ▣ **DM**: Diebold–Mariano (test)
- ▣ **EA**: Euro Area
- ▣ **ECB**: European Central Bank
- ▣ **ENTSO-E**: European Network of Transmission System Operators for Electricity
- ▣ **EODHD**: EOD Historical Data (market-data provider)
- ▣ **ES**: Expected Shortfall
- ▣ **FRED**: Federal Reserve Economic Data
- ▣ **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ **GARCH-N**: GARCH with Normal innovations
- ▣ **GDP**: Gross Domestic Product
- ▣ **GW**: Giacomini–White (test of conditional predictive ability)
- ▣ **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ **HICP**: Harmonised Index of Consumer Prices
- ▣ **HLN**: Harvey–Leybourne–Newbold (small-sample correction of the DM test)
- ▣ **INS**: Institutul Național de Statistică (the National Institute of Statistics (Romania))

Acronyms used in this material (2/2)

- ▣ **ISE:** (Fraunhofer) Institute for Solar Energy Systems
- ▣ **LLM:** Large Language Model
- ▣ **LR:** Likelihood Ratio (test)
- ▣ **MA:** Moving Average
- ▣ **MAE:** Mean Absolute Error
- ▣ **MAPE:** Mean Absolute Percentage Error
- ▣ **MASE:** Mean Absolute Scaled Error
- ▣ **MCS:** Model Confidence Set
- ▣ **MSE:** Mean Squared Error
- ▣ **MSPE:** Mean Squared Prediction Error
- ▣ **MW:** MegaWatt
- ▣ **MZ:** Mincer–Zarnowitz (regression)
- ▣ **OLS:** Ordinary Least Squares
- ▣ **PIT:** Probability Integral Transform
- ▣ **QLIKE:** Quasi-LIKElihood loss
- ▣ **RC:** White's Reality Check (for data snooping)
- ▣ **RMSE:** Root Mean Squared Error
- ▣ **RTDSM:** Real-Time Data Set for Macroeconomists (Philadelphia Fed)
- ▣ **RW:** Random Walk
- ▣ **SE:** Standard Error
- ▣ **SPA:** Superior Predictive Ability (test)
- ▣ **SPF:** Survey of Professional Forecasters (Federal Reserve Bank of Philadelphia)
- ▣ **TSA:** Time Series Analysis
- ▣ **UIP:** Uncovered Interest Parity
- ▣ **US:** United States
- ▣ **VAR:** Vector AutoRegression

Today's question and route

- ▣ **Question:** when is one forecast *really* better than another, and what do we gain by combining them?
 - ▶ “better” needs a loss or a score, a reference distribution of the comparison and an honest evaluation design
- ▣ **Route** of the chapter
 - ▶ point forecasts: which functional a loss function rewards (consistency, Bregman losses)
 - ▶ probabilistic forecasts: calibration, sharpness, PIT, proper scoring rules, elicibility
 - ▶ comparing forecasts: DM with HAC and small-sample corrections, GW, Clark–West, MZ, encompassing, RC, SPA, MCS
 - ▶ combination: Bates–Granger, the combination puzzle, density pools, the M4 and M5 lessons; real-time data
- ▣ We build on [TSA, Chapter 0](#) and [TSA, Chapter 4](#) (MAE, RMSE, MASE, rolling-origin evaluation, the basic DM test) and on [Chapter 0](#) (HAC, bootstrap); Seminar 1 comes before this lecture

Learning outcomes

- ▣ Derive the optimal point forecast under squared, absolute, pinball and asymmetric losses, and recognise the losses consistent for the mean (Bregman)
- ▣ Assess the calibration of density forecasts with PIT histograms and tests, and rank them with proper scores (log score, CRPS, interval and energy scores)
- ▣ Explain elicibility and why VaR is elicitable while ES is elicitable only jointly with VaR
- ▣ Test equal predictive ability correctly: DM–HLN with HAC variance, GW, Clark–West for nested models, MZ, encompassing, SPA and the MCS
- ▣ Combine point and density forecasts and explain the forecast combination puzzle; evaluate forecasts with real-time data

Reading, data and tools

- ▣ Surveys: [Petropoulos et al. \(2022\)](#) (open access), [Gneiting and Katzfuss \(2014\)](#), [Elliott and Timmermann \(2008\)](#), [Timmermann \(2006\)](#), [Wang et al. \(2023\)](#)
 - ▶ textbooks: [Hamilton \(1994\)](#), Ch. 4 (forecasting); [Kilian and Lütkepohl \(2017\)](#); bridge: [Huang and Petukhina \(2022\)](#)
- ▣ Python Quantlets of this chapter: [Quantlets/Ch_01](#)
 - ▶ DM–HLN, GW, Clark–West, MZ, encompassing, Berkowitz, CRPS and the MCS written out in `numpy`; Hansen’s SPA from the `arch` package
- ▣ Lecture notebook: [open in Google Colab](#)

Data used in this chapter

Series	Source	Use
HICP inflation, Romania (y/y, monthly)	Eurostat (prc_hicp_minr)	DM, GW
CPI and unemployment, US (quarterly)	FRED (CPIAUCSL, UNRATE)	Atkeson–Ohanian
EUR/RON and 3-month rates RO, EA (monthly)	BNR reference rate; Eurostat (irt_st_m)	Clark–West, SPA
S&P 500, daily close	EODHD	PIT, scores
Electricity load, Romania (hourly)	Energy-Charts (Fraunhofer ISE), ENTSO-E data	pinball, MCS
Individual forecasts, US SPF	Federal Reserve Bank of Philadelphia	combination
US real GDP, all releases	Philadelphia Fed, RTDSM	real time

- All sources are public and need no account or key; daily and monthly data up to September 2026
- HICP: harmonised index of consumer prices; CPI: consumer price index; SPF: Survey of Professional Forecasters; RTDSM: Real-Time Data Set for Macroeconomists

From weather forecasts to forecast verification



Robert FitzRoy (1805–1865)

Portrait: Charles Hemus (c. 1910); public domain;

Wikimedia Commons



Lewis Fry Richardson (1881–1953)

Photo: NOAA; public domain; Wikimedia Commons

- 1861: FitzRoy, head of the new Meteorological Office, publishes daily “forecasts” in The Times
 - ▶ he coined the word in its modern sense; the forecasts were often wrong and publicly mocked
- 1922: Richardson computes a weather forecast by numerical integration of the equations of motion, by hand
- 1950: Brier (1950) proposes a score for probability forecasts of rain; meteorology leads forecast verification for the next fifty years
 - ▶ economics adopts the same tools late: PIT Diebold, Gunther and Tay (1998), proper scores Gneiting and Raftery (2007), calibration Gneiting, Balabdaoui and Raftery (2007)
- Lesson: a forecast is only as useful as the way we evaluate it

Setting and notation

- Target Y_{t+h} , information set $\mathcal{F}_t$, horizon h ; a **point forecast** $\hat{y}_{t+h|t}$ is $\mathcal{F}_t$ -measurable
 - ▶ $\mathcal{F}_t$ -measurable: computed only from information available at t
 - ▶ a **probabilistic forecast** is a predictive distribution $F_{t+h|t}$ (density, quantiles, interval, ensemble)
- Out-of-sample design: R observations to estimate, P forecasts to evaluate, $R + P + h - 1 = T$
 - ▶ schemes: **fixed** (estimate once), **rolling** (last R), **recursive** (all past); TSA, Chapter 4 called this rolling-origin evaluation
 - ▶ T : the total sample; the last forecast, made at $R + P - 1$, targets T ; the scheme matters for inference: West (1996) versus Giacomini and White (2006), Section 4
- Known from TSA, Chapter 0: MAE, RMSE, MAPE and the scale-free MASE Hyndman and Koehler (2006)
 - ▶ $\text{MAE} = P^{-1} \sum |e_t|$, $\text{RMSE} = (P^{-1} \sum e_t^2)^{1/2}$, $\text{MAPE} = 100P^{-1} \sum |e_t/y_t|$; MASE: the MAE divided by the in-sample MAE of the naive forecast
 - ▶ here the question is different: *which* loss, and what does it reward?

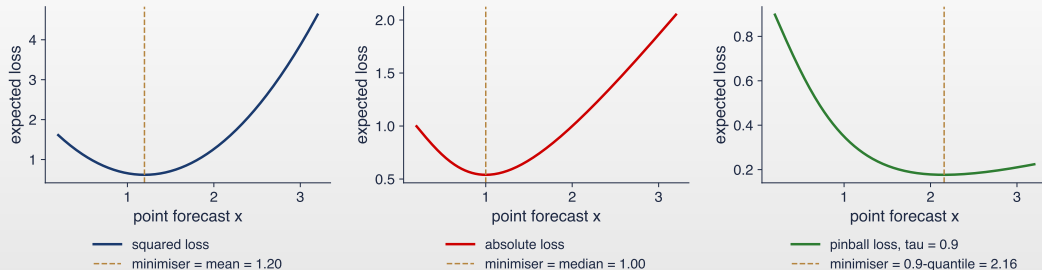
The optimal point forecast and consistency (1/2)

- Given a loss $L(x, y)$ and a predictive distribution F , the **optimal** (Bayes) forecast minimises the expected loss:
 - ▶ $x^* = \arg \min_x \mathbb{E}_F[L(x, Y)]$
 - ▶ x : the point forecast; y : the outcome; $L(x, y) \geq 0$: the penalty; $\mathbb{E}_F$: the expectation when $Y \sim F$
- A **functional** $T(F)$ maps a distribution to a number: the mean, the median, a quantile
 - ▶ L is **consistent** for T if $\mathbb{E}_F L(T(F), Y) \leq \mathbb{E}_F L(x, Y)$ for all x and F [Gneiting \(2011\)](#)
 - ▶ i.e. reporting $T(F)$ is optimal whatever F is; **strictly** consistent: equality only at $x = T(F)$

The optimal point forecast and consistency (2/2)

- Three classical cases, with the forecast error $e = y - x$; set the derivative of the expected loss to zero
 - ▶ squared e^2 : $\frac{d}{dx} \mathbb{E}(Y - x)^2 = -2\mathbb{E}(Y - x) = 0 \Rightarrow x^* = \mathbb{E}Y$ (the mean)
 - ▶ absolute $|e|$: $\frac{d}{dx} \mathbb{E}|Y - x| = F(x) - (1 - F(x)) = 0 \Rightarrow F(x^*) = 1/2$ (the median)
 - ▶ pinball $\rho_\tau(e) = (\tau - 1\{e < 0\})e$: $F(x) - \tau = 0 \Rightarrow x^* = F^{-1}(\tau)$ (the τ -quantile)
- Notation of the pinball loss
 - ▶ $\tau \in (0, 1)$: the quantile level; $1\{e < 0\}$: 1 if the forecast is too high, 0 otherwise; F^{-1} : the quantile function
 - ▶ under-forecasts cost $\tau|e|$, over-forecasts $(1 - \tau)|e|$: a high τ pushes the forecast up
- Consequence: announce the loss *before* asking for a forecast; a median forecast judged by RMSE is judged unfairly [Gneiting \(2011\)](#)

Three losses, three optimal forecasts



□ Y log-normal with $\ln Y \sim N(0, 0.6^2)$; expected loss as a function of the point forecast x (simulation, 2×10^5 draws)

ATS_ch1_loss_functions

Interpreting the three minimisers

- Mean 1.20, median 1, 0.9-quantile 2.16; three different “best” forecasts of the same variable
 - ▶ for a right-skewed target (wages, loads, prices, volatility) the gap is large
- The pinball curve is flat near its minimum: quantile forecasts are hard to separate with few observations
- A forecaster who knows the loss reports the matching functional; ranking forecasts by another loss rewards the wrong behaviour

Bregman losses: all losses consistent for the mean (1/2)

- **Theorem Savage (1971), Gneiting (2011):** under regularity, L is consistent for the mean iff it is a **Bregman** loss
 - ▶ $L(x, y) = \phi(y) - \phi(x) - \phi'(x)(y - x)$
 - ▶ ϕ : a convex function, ϕ' its derivative; L is the gap between $\phi(y)$ and the tangent of ϕ at x , so $L \geq 0$ (up to terms in y only)
- Two members of the family
 - ▶ $\phi(x) = x^2$: the squared error $(y - x)^2$
 - ▶ $\phi(x) = -\ln x$: **QLIKE** $y/x - \ln(y/x) - 1$, for a variance forecast $x > 0$ and a variance proxy $y > 0$ ([Chapter 8](#))
 - ▶ the conditional mean is optimal for every Bregman loss [Banerjee et al. \(2005\)](#); proof for the quantile in the

▶ [Appendix](#)

Bregman losses: all losses consistent for the mean (2/2)

- Proof sketch, with $\mu = \mathbb{E}Y$:
 - ▶ $\mathbb{E}L(x, Y) - \mathbb{E}L(\mu, Y) = \phi(\mu) - \phi(x) - \phi'(x)(\mu - x) \geq 0$
 - ▶ the linear term vanishes because $\mathbb{E}(Y - \mu) = 0$; the inequality is convexity of ϕ
- Correctly specified forecasts rank the same under every Bregman loss; misspecified ones need not [Patton \(2020\)](#)
 - ▶ Murphy diagrams [Ehm et al. \(2016\)](#) plot the score over the whole family: one forecast dominates only if its curve is lower everywhere
 - ▶ robust proxies for volatility: only some losses keep the ranking with a noisy proxy [Patton \(2011\)](#)

Asymmetric loss

- **LinEx** loss $L(e) = b[\exp(ae) - ae - 1]$, $e = y - x$: linear on one side, exponential on the other
 - ▶ $b > 0$: a scale; a : the asymmetry; $a > 0$ penalises under-forecasts ($e > 0$) exponentially, over-forecasts about linearly
 - ▶ optimal forecast $x^* = a^{-1} \ln \mathbb{E} e^{aY}$; with $Y \mid \mathcal{F}_t \sim N(\mu_t, \sigma_t^2)$ (conditional mean μ_t , variance σ_t^2):
 $x^* = \mu_t + a\sigma_t^2/2$ [Christoffersen and Diebold \(1997\)](#)
 - ▶ the optimal forecast is **biased** and its bias moves with the conditional variance
- Consequences for evaluation
 - ▶ a rational forecaster under asymmetric loss fails the MZ unbiasedness test (Section 4)
 - ▶ the pinball loss is the piecewise-linear member: the optimal forecast is a quantile
- Examples: a grid operator (under-forecasting load is costlier than over-forecasting), a central bank with asymmetric inflation costs, the capital buffer of a bank ([Chapter 9](#))

Recap: Point forecasts and loss functions

- A loss defines the target functional: squared $\rightarrow$ mean, absolute $\rightarrow$ median, pinball $\rightarrow$ quantile
- Bregman losses are exactly those consistent for the mean; QLIKE is one of them
- Under asymmetric loss the optimal forecast is biased by design
- With misspecified models the ranking can depend on the loss: fix it in advance

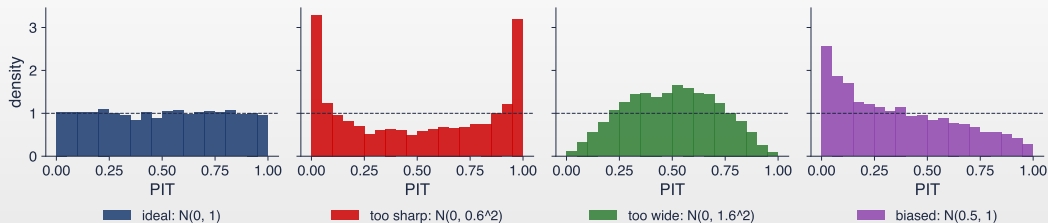
Maximise sharpness subject to calibration

- **Calibration:** statistical consistency between the forecasts F_t and the outcomes y_t (a joint property) [Gneiting, Balabdaoui and Raftery \(2007\)](#)
 - ▶ **probabilistic** calibration: the PIT (probability integral transform) $u_t = F_t(y_t)$, the forecast probability of an outcome below y_t , is uniform on average
 - ▶ **marginal** calibration: $\frac{1}{T} \sum_t F_t(x) \rightarrow G(x)$ for every x , with G the limit of the empirical distribution of y_t
- **Sharpness:** concentration of the predictive distributions; a property of the forecasts only
 - ▶ measured by the average width of central intervals or the average predictive variance
- Paradigm [Gneiting, Balabdaoui and Raftery \(2007\)](#): among calibrated forecasts, prefer the sharpest
 - ▶ the climatological forecast (the unconditional distribution) is calibrated but not sharp
 - ▶ proper scores (next section) reward both at once

The probability integral transform

- **Theorem** Dawid (1984), Diebold, Gunther and Tay (1998): if F_t is the true conditional distribution of Y_t given $\mathcal{F}_{t-1}$ and continuous, then $u_t = F_t(y_t)$ are i.i.d. $U(0, 1)$
 - ▶ uniform: for every $v \in (0, 1)$, $P(u_t \leq v \mid \mathcal{F}_{t-1}) = P(Y_t \leq F_t^{-1}(v) \mid \mathcal{F}_{t-1}) = v$, whatever the past
 - ▶ independent: a conditional distribution that does not depend on the past implies independence of the sequence
- For $h > 1$ the PITs are uniform but serially dependent up to lag $h - 1$ (overlapping horizons)
 - ▶ tests must allow this dependence Rossi and Sekhposyan (2019)
- Uniformity is necessary, not sufficient: Hamill (2001) and Gneiting, Balabdaoui and Raftery (2007) build uncalibrated forecasters whose PIT is uniform on average
 - ▶ check PIT conditionally: by sub-sample, by regime, by horizon

PIT histogram shapes



□ $Y \sim N(0, 1)$, 5000 draws; four Normal forecasts; dashed line: the uniform density

ATS_ch1_calibration_scores

Interpreting the PIT shapes

- ▣ **U-shape**: too many outcomes in both tails (32% outside the central 90%, against 10%): the forecast is too sharp, intervals too narrow
- ▣ **Hump**: too few outcomes in the tails (1.0%): the forecast is too wide, the intervals waste information
- ▣ **Slope**: a biased location; here the forecast mean is too high, so PIT values cluster near 0
- ▣ CRPS (lower is better): ideal 0.567, too sharp 0.595, too wide 0.604, biased 0.637: the proper score ranks the ideal forecast first

Testing calibration (1/2)

- ▣ **Berkowitz test** [Berkowitz \(2001\)](#): if the PITs are i.i.d. uniform, $z_t = \Phi^{-1}(u_t)$ is i.i.d. $N(0, 1)$
 - ▶ Φ^{-1} : the quantile function of the standard Normal distribution
- ▣ Fit a Gaussian AR(1) to z_t and test its parameters:
 - ▶ $z_t = \mu + \rho(z_{t-1} - \mu) + \varepsilon_t$, $\varepsilon_t \sim N(0, \sigma^2(1 - \rho^2))$
 - ▶ μ : the mean of z_t ; σ^2 : its variance; ρ : its first-order autocorrelation; calibration means $\mu = 0$, $\sigma = 1$, $\rho = 0$
 - ▶ LR (likelihood ratio) statistic $2(\ell_1 - \ell_0) \sim \chi^2(3)$ under H_0 ; ℓ_1, ℓ_0 : maximised log-likelihoods without and with the restrictions
 - ▶ the transformation to z makes the tails visible; the test checks only the first two moments and AR(1) dependence

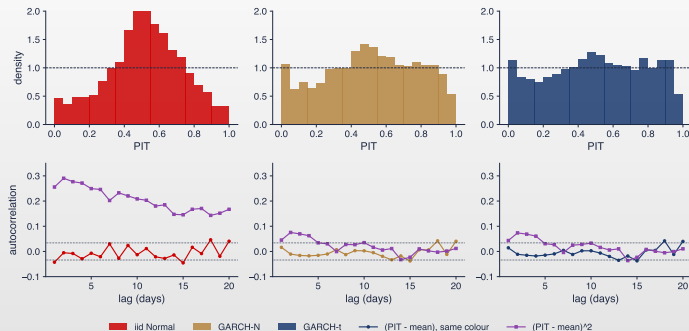
Testing calibration (2/2)

- ▣ **Correlograms** of $(u_t - \bar{u})^k$, $k = 1, 2$ [Diebold, Gunther and Tay \(1998\)](#): dependence in level and in volatility
 - ▶ $\bar{u}$: the mean PIT; Ljung–Box on $(u_t - \bar{u})^2$ detects missing volatility dynamics
- ▣ Parameter estimation changes the null distribution of PIT tests
 - ▶ with P/R small (few forecasts relative to the estimation sample) it can be ignored, otherwise use [Rossi and Sekhposyan \(2019\)](#)
- ▣ Interval forecasts: coverage and independence of the hits [Christoffersen \(1998\)](#)
 - ▶ hit: an outcome outside the interval; for VaR this becomes backtesting ([Chapter 9](#))

Case study: Diebold, Gunther and Tay (1998)

- The paper: daily S&P 500 returns; parameters estimated once on the first part of the sample, density forecasts evaluated on the second part with PIT histograms and correlograms of $(u - \bar{u})^k$ Diebold, Gunther and Tay (1998)
 - ▶ their models: i.i.d. Normal; MA(1)-GARCH(1,1) with Normal and with Student- t innovations Bollerslev (1987)
- Our replication: the same three models, estimated by maximum likelihood on 2000–2012, fixed parameters, one-day density forecasts for 2013–2026 (3 448 days)
 - ▶ model: $r_t = c + \varepsilon_t + \theta\varepsilon_{t-1}$, $\varepsilon_t = \sigma_t z_t$, $\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$; z_t Normal or Student- t with ν degrees of freedom
 - ▶ GARCH- t : $\hat{\theta} = -0.061$, $\hat{\alpha} + \hat{\beta} = 0.996$ (the persistence of volatility), $\hat{\nu} = 8.0$ (fat tails)
- Question: which density forecast is calibrated, and do the proper scores agree with the PIT diagnosis?

PIT diagnostics of three S&P 500 density forecasts



Top: PIT histograms (20 bins); bottom: autocorrelations of $(u - \bar{u})$ and $(u - \bar{u})^2$ with $\pm 2/\sqrt{T}$ bands

Interpreting the S&P 500 diagnostics

- i.i.d. Normal: a hump and strong autocorrelation of $(u - \bar{u})^2$: the forecast ignores volatility clustering; Berkowitz LR 428.8 ($p < 0.001$)
 - ▶ too wide in calm periods, too narrow in turbulent ones: the two errors average into a hump
- GARCH removes most of the dependence; the t version has the better left tail
 - ▶ Berkowitz: GARCH-N LR 9.3 (p 0.026), GARCH- t LR 10.0 (p 0.019); estimated σ_z $0.97 < 1$
 - ▶ outside the central 90%: 8.4% of days: with parameters frozen in 2012 the forecasts are slightly too wide after 2013
- In line with [Diebold, Gunther and Tay \(1998\)](#): conditional heteroskedasticity is essential, fat tails help; a small residual miscalibration remains

Recap: Calibration

- ▣ Calibration is a joint property of forecasts and outcomes; sharpness is a property of the forecasts
- ▣ Correct one-step densities give i.i.d. uniform PITs; for $h > 1$ they are dependent
- ▣ U-shape: too sharp; hump: too wide; slope: bias; correlogram of $(u - \bar{u})^2$: missing volatility dynamics
- ▣ The S&P 500: GARCH- t is close to calibrated, the i.i.d. Normal is not

Proper scoring rules

- A **scoring rule** $S(F, y)$ assigns a penalty to the forecast F when y occurs (negatively oriented: lower is better)
 - ▶ **proper**: $\mathbb{E}_G S(G, Y) \leq \mathbb{E}_G S(F, Y)$ for all F, G [Gneiting and Raftery \(2007\)](#)
 - ▶ G : the distribution the forecaster believes; $\mathbb{E}_G$: the expectation under G ; reporting the true belief is optimal
 - ▶ **strictly proper**: equality only if $F = G$; then the score cannot be manipulated
- Proper scores reward calibration and sharpness together: the expected score decomposes into uncertainty, minus resolution, plus reliability
 - ▶ the Brier score $(p - 1\{\text{event}\})^2$, with p the forecast probability of a binary event, is the oldest example [Brier \(1950\)](#)
- A proper score of a probabilistic forecast reduces to a consistent loss of a point forecast when F is a point mass (CRPS $\rightarrow$ absolute error)

The logarithmic score

- $\text{LogS}(f, y) = -\ln f(y)$ [Good \(1952\)](#): the only proper score that is **local** (depends on f only at y , up to equivalence)
 - ▶ f : the forecast density; g : the true density; properness:
$$\mathbb{E}_g[\text{LogS}(f, Y)] - \mathbb{E}_g[\text{LogS}(g, Y)] = \int g \ln(g/f) = \text{KL}(g\|f) \geq 0$$
 - ▶ KL: the Kullback–Leibler divergence, nonnegative and zero only if $f = g$ (Gibbs inequality)
- Links: average log score = minus the out-of-sample log-likelihood; the difference of two log scores is a predictive likelihood ratio
 - ▶ Bayes factors and the predictive model choice of [Geweke and Amisano \(2011\)](#) are built on it
- Weaknesses: infinite if $f(y) = 0$; very sensitive to tail events; needs a density
 - ▶ cannot score ensembles or quantile forecasts directly

The continuous ranked probability score (1/2)

- Definition [Matheson and Winkler \(1976\)](#): the squared distance between the forecast distribution function and the step function of the outcome
 - ▶ $\text{CRPS}(F, y) = \int_{-\infty}^{\infty} (F(z) - 1\{y \leq z\})^2 dz$
 - ▶ z : a threshold; $1\{y \leq z\}$: the “distribution function” of the outcome, 0 below y and 1 above; units: those of y
- Kernel form [Gneiting and Raftery \(2007\)](#) (proof in the [Appendix](#)):
 - ▶ $\text{CRPS}(F, y) = \mathbb{E}_F |X - y| - \frac{1}{2} \mathbb{E}_F |X - X'|$, $X, X' \sim F$ independent
 - ▶ first term: accuracy; second: a reward for spread that stops the score from favouring point masses
 - ▶ for a point forecast (F a point mass at x) it reduces to the absolute error $|y - x|$

The continuous ranked probability score (2/2)

- Closed form for $F = N(\mu, \sigma^2)$, with $z = (y - \mu)/\sigma$:
 - ▶ $\text{CRPS} = \sigma [z(2\Phi(z) - 1) + 2\varphi(z) - 1/\sqrt{\pi}]$
 - ▶ Φ, φ : the distribution and density functions of $N(0, 1)$; a closed form exists also for Student t (Quantlet)
 - ▶ ensembles: plug in the empirical distribution of the members
- Decompositions:
 - ▶ $\text{CRPS} = \int \text{BS}_z dz = 2 \int_0^1 \rho_\tau(y - F^{-1}(\tau)) d\tau$
 - ▶ BS_z : the Brier score of the event $\{y \leq z\}$; ρ_τ : the pinball loss at level τ
 - ▶ so the average pinball over the 99 percentiles (GEFCom2014) approximates $\text{CRPS}/2$ [Hong et al. \(2016\)](#)

Quantile, interval and multivariate scores (1/2)

- ▣ **Quantile score:** $\rho_{\tau}(y - q)$ for a forecast q of the τ -quantile, strictly consistent [Gneiting \(2011\)](#)
 - ▶ any increasing g gives another consistent score: $(1\{y \leq q\} - \tau)(g(q) - g(y))$
- ▣ **Interval score** of a central $(1 - \alpha)$ interval $[l, u]$ [Gneiting and Raftery \(2007\)](#)
 - ▶ $IS_{\alpha} = (u - l) + \frac{2}{\alpha}(l - y)1\{y < l\} + \frac{2}{\alpha}(y - u)1\{y > u\}$
 - ▶ l, u : the lower and upper limits; width plus a penalty $2/\alpha$ per unit of miss
 - ▶ it equals $\frac{2}{\alpha}[\rho_{\alpha/2}(y - l) + \rho_{1-\alpha/2}(y - u)]$; coverage alone is not a score

Quantile, interval and multivariate scores (2/2)

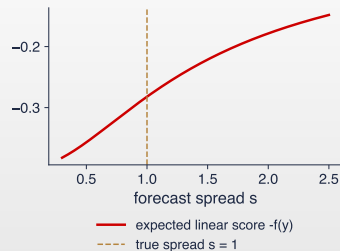
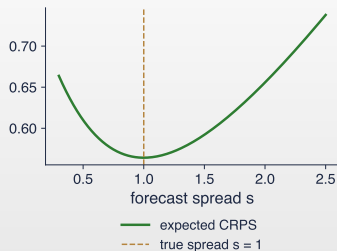
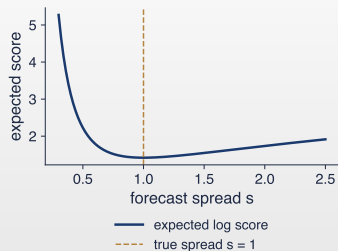
▣ **Energy score** for vectors, the multivariate CRPS:

- ▶ $ES(F, y) = \mathbb{E}\|X - y\| - \frac{1}{2}\mathbb{E}\|X - X'\|$
- ▶ y : the outcome vector; X, X' : independent draws from the forecast F ; $\|\cdot\|$: the Euclidean norm
- ▶ weak at detecting wrong correlations; the **variogram score** [Scheuerer and Hamill \(2015\)](#) targets the dependence structure (24 hourly loads, a VAR path)
- ▶ $VS_p = \sum_{i,j} w_{ij} (|y_i - y_j|^p - \mathbb{E}|X_i - X_j|^p)^2$: compares observed and forecast differences between components; $w_{ij} \geq 0$: weights, p : usually 0.5

▣ Tails: threshold-weighted CRPS stays proper [Gneiting and Ranjan \(2011\)](#)

- ▶ $\int (F(z) - 1\{y \leq z\})^2 w(z) dz$
- ▶ $w(z) \geq 0$: a weight that emphasises the thresholds of interest (e.g. the left tail); the weighted likelihood ratio of [Amisano and Giacomini \(2007\)](#) does not stay proper [Gneiting and Ranjan \(2011\)](#)

Proper and improper scores



- Truth $Y \sim N(0, 1)$; forecast $N(0, s^2)$; expected score as a function of the spread s (closed forms)
- Log score and CRPS: minimum at $s = 1$; linear score $-f(y)$: $\mathbb{E}f(Y) = [2\pi(1 + s^2)]^{-1/2}$ grows as $s \rightarrow 0$: it rewards overconfidence

Interpreting the expected scores

- The log score and the CRPS are minimised by the true spread: a forecaster maximises the expected reward by reporting what she believes
- The log score punishes overconfidence much more than excess spread (steep left branch); the CRPS is more symmetric and bounded by the absolute error
- The linear score is maximised by a point mass: it would reward a forecaster who always reports a spike, so it must not be used for evaluation

Elicibility

- A functional T is **elicitable** if a strictly consistent loss exists for it [Gneiting \(2011\)](#)
 - ▶ mean, quantiles, expectiles, ratios of expectations: elicitable
 - ▶ only elicitable functionals can be compared and ranked meaningfully with point forecasts
- **Necessary condition** (Osband): convex level sets: $T(F_0) = T(F_1) = t \Rightarrow T(\lambda F_0 + (1 - \lambda)F_1) = t$
 - ▶ F_0, F_1 : two distributions with the same value t ; $\lambda \in [0, 1]$: the mixing weight
 - ▶ the **variance** fails it, so it is not elicitable; the pair (mean, second moment) is
- Risk measures with $\text{VaR}_\alpha(X) = -q_\alpha(X)$, q_α the α -quantile of the return X (VaR 1%: $\alpha = 0.01$)
 - ▶ VaR is a quantile: elicitable by the pinball loss
 - ▶ ES (expected shortfall, the mean loss beyond VaR) has non-convex level sets: **not elicitable** alone [Weber \(2006\)](#), [Gneiting \(2011\)](#); (VaR, ES) is **jointly** elicitable [Fissler and Ziegel \(2016\)](#)
 - ▶ the Fissler–Ziegel scores and the backtests built on them: [Chapter 9](#)

Scores of the S&P 500 density forecasts

Model	Log score	CRPS	Outside 90%	Berkowitz p
i.i.d. Normal	1.531	0.564	3.9%	< 0.001
MA(1)-GARCH(1,1)-N	1.265	0.509	8.0%	0.026
MA(1)-GARCH(1,1)- t	1.233	0.507	8.4%	0.019

- Score differences tested with DM on the daily scores ([Amisano and Giacomini \(2007\)](#), unweighted)
 - GARCH- t against GARCH-N: HLN -5.9 (log score), -6.4 (CRPS); GARCH-N against i.i.d.: -12.7 and -21.5
- Both scores rank the models in the same order as the PIT diagnosis; the log score separates the tails more sharply
- Ranking and calibration are different questions: the best-scoring model can still fail a calibration test

Recap: Proper scores and elicibility

- Proper scores make honesty optimal; strictly proper scores cannot be gamed
- LogS: local, tail-sensitive; CRPS: robust, works for ensembles, equals twice the integrated pinball loss
- Interval score for intervals, energy and variogram scores for vectors
- Elicitable: mean, quantile (VaR); not elicitable: variance, ES alone; (VaR, ES) jointly

Two questions about predictive ability

- ▣ **Population** West (1996): are the *models*, at their pseudo-true parameters, equally accurate?
 - ▶ estimation error enters the asymptotic variance unless $P/R \rightarrow 0$ or the loss is the one used in estimation
- ▣ **Finite sample** Giacomini and White (2006): are the *forecasting methods*, with their estimation windows, equally accurate?
 - ▶ rolling window of fixed size R : estimation noise is part of the method; nested models allowed
- ▣ **Diebold–Mariano** Diebold and Mariano (1995) treats the forecasts as given (“primitives”): it compares forecasts, not models Diebold (2015)
 - ▶ using DM to compare models with estimated parameters needs the West or GW framework

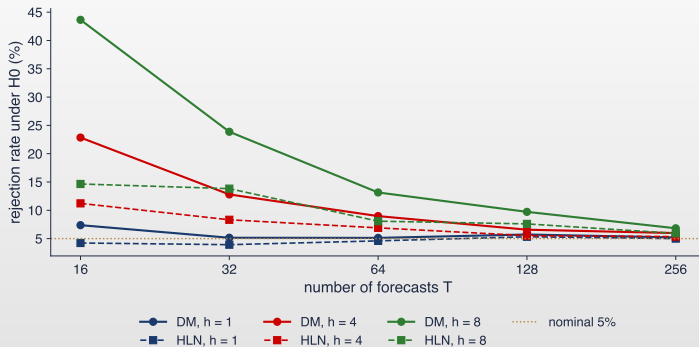
The Diebold–Mariano test with HAC variance (1/2)

- ▣ Loss differential $d_t = L(e_{1t}) - L(e_{2t})$; $H_0: \mathbb{E}d_t = 0$ (equal accuracy)
 - ▶ e_{it} : the error of forecast i at t ; $d_t > 0$: forecast 2 did better at t ; d_t assumed covariance stationary
- ▣ The statistic: the mean differential over its HAC standard error
 - ▶ $DM = \bar{d} / \sqrt{\hat{\sigma}_{LR}^2 / P} \rightarrow N(0, 1)$, $\sigma_{LR}^2 = \gamma_0 + 2 \sum_{k \geq 1} \gamma_k$
 - ▶ $\bar{d}$: the mean of d_t over the P forecasts; γ_k : the autocovariances of d_t ; σ_{LR}^2 : its long-run variance (Chapter 0)
 - ▶ a large positive DM: forecast 2 is more accurate; reject at 5% if $|DM| > 1.96$

The Diebold–Mariano test with HAC variance (2/2)

- Optimal h -step errors are at most $\text{MA}(h - 1)$, so DM truncate at $h - 1$ lags with a **rectangular** kernel
 - ▶ the rectangular estimate can be **negative**; Newey–West (Bartlett) weights keep it positive [Newey and West \(1987\)](#), at the cost of downward bias
 - ▶ suboptimal forecasts have errors with longer memory: use a data-driven bandwidth ([Chapter 0](#))
- **HLN correction** [Harvey, Leybourne and Newbold \(1997\)](#): a small-sample rescaling of DM
 - ▶ $\text{DM}^* = \text{DM} \sqrt{(P + 1 - 2h + h(h - 1)/P)/P}$, compared with t_{P-1}
 - ▶ the factor (below 1) removes the bias of the variance estimator for $\text{MA}(h - 1)$ errors; t_{P-1} : Student t with $P - 1$ degrees of freedom, for small P

Size of DM and HLN in small samples



Two independent $MA(h-1)$ error series with equal variance, squared loss; share of rejections at 5% in 4 000 replications

Interpreting the size experiment

- $h = 1$: both tests are close to 5%; DM over-rejects only at $P = 16$ (7.4%)
- $h = 8, P = 16$: DM rejects a true null 43.6% of the time, HLN 14.6%; at $P = 64$: 13.2% and 8.1%
 - ▶ the HAC variance with few observations and many lags is noisy and biased down
- Rule: at multi-step horizons with fewer than about 100 forecasts, use HLN and treat p -values near 5% as inconclusive; fixed- b critical values ([Chapter 0](#)) are an alternative

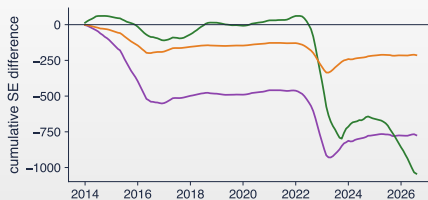
Romanian inflation one year ahead: the design

- ▣ Target: HICP inflation, y/y, 12 months ahead; forecast origins from January 2013, targets January 2014 to August 2026, $P = 152$ forecasts
 - ▶ HICP inflation in August 2026: 6.3%
- ▣ Forecasts
 - ▶ **no change**: $\hat{\pi}_{t+12} = \pi_t$; **direct AR(3)**: π_{t+12} on $1, \pi_t, \pi_{t-1}, \pi_{t-2}$, OLS on a 10-year rolling window
 - ▶ **target**: the BNR inflation target, 2.5%; **average** of AR(3) and no change
- ▣ Overlapping 12-month targets: errors are serially correlated up to lag 11; DM with $h = 12$

Romanian inflation forecasts one year ahead



— HICP inflation, Romania (y/y, %)
 — no change (made 12 months earlier)
 — direct AR(3), rolling 10 years
 - - - BNR target 2.5%



— AR(3) against no change
 — target against no change
 — average of AR and no change against no change

- Left: inflation and the forecasts made 12 months earlier; right: cumulative sum of $e_{nc,t}^2 - e_{m,t}^2$ (falling: model m loses to no change)

Interpreting the Romanian inflation comparison

Forecast	RMSE	DM	HLN	p (HLN)	p , abs. loss
no change	3.57	—	—	—	—
AR(3)	4.23	1.43	1.32	0.188	0.381
BNR target	4.43	1.30	1.21	0.230	0.319
average	3.76	0.96	0.88	0.378	0.795

- Every model has a higher RMSE than no change, yet no difference is significant: with $P = 152$ overlapping targets the effective sample is small
- Newey–West instead of the rectangular kernel: HLN 1.47 for AR(3); the conclusion does not depend on the kernel
- Relative RMSE of AR(3): 1.44 before July 2021, 1.10 after: the ranking is not stable over time [Giacomini and Rossi \(2010\)](#)

Conditional predictive ability: Giacomini–White (1/2)

- $H_0: \mathbb{E}[d_{t+h} | \mathcal{G}_t] = 0$ Giacomini and White (2006)
 - ▶ $\mathcal{G}_t$: the information at the forecast origin; H_0 : no instrument $h_t \in \mathcal{G}_t$ predicts which method wins
 - ▶ h_t : a $q \times 1$ vector of instruments (not the horizon h), e.g. a constant and the last loss differential
- The statistic: a Wald test that the instruments do not predict d_{t+h}
 - ▶ $GW = P \bar{Z}' \hat{\Omega}^{-1} \bar{Z} \rightarrow \chi^2(q)$, $Z_t = h_t d_{t+h}$
 - ▶ $\bar{Z}$: the mean of Z_t ; $\hat{\Omega}$: its HAC covariance matrix with $h - 1$ lags; $\chi^2(q)$: chi-squared with q degrees of freedom
 - ▶ $h_t = 1$ gives the unconditional test; adding d_t or a state variable tests whether the past predicts the winner

Conditional predictive ability: Giacomini–White (2/2)

- ▣ Valid with a **rolling** window of fixed size, nested or non-nested models, any loss; invalid with a recursive window
 - ▶ a rejection suggests a decision rule: use model 1 when $\hat{\delta}'h_t > 0$, with $\hat{\delta}$ the regression of d_{t+h} on h_t
- ▣ Romanian inflation, AR(3) against no change, $h_t = (1, d_t, \pi_t - 2.5)$
 - ▶ $\pi_t - 2.5$: the distance of inflation from the BNR target
 - ▶ $GW = 3.19$, p 0.363: neither the recent loss nor the distance from the target predicts the winner

Nested models: the failure of DM and the Clark–West test (1/2)

- Model 1 (small, e.g. the random walk) is nested in model 2; under H_0 the extra coefficients are zero
 - ▶ MSPE: the mean squared prediction error; population MSPEs are equal, and d_t is degenerate
 - ▶ so the DM statistic is not asymptotically Normal [Clark and McCracken \(2001\)](#)
- In finite samples model 2 estimates zero coefficients with noise
 - ▶ its MSPE is **larger** than that of model 1: DM is undersized (rejects too rarely) and has little power

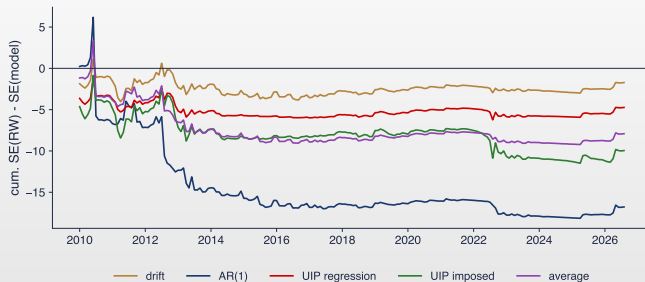
Nested models: the failure of DM and the Clark–West test (2/2)

- **Clark–West** [Clark and West \(2007\)](#): adjust the larger model for the noise it adds
 - ▶ $f_t = e_{1t}^2 - [e_{2t}^2 - (\hat{y}_{1t} - \hat{y}_{2t})^2]$
 - ▶ e_{1t}, e_{2t} : the errors of the small and large model; $\hat{y}_{1t}, \hat{y}_{2t}$: their forecasts; the squared gap between forecasts is the noise added by model 2
 - ▶ one-sided t test of $\mathbb{E}f_t = 0$ against $\mathbb{E}f_t > 0$ (model 2 better) with Normal critical values
- Why the adjustment works
 - ▶ under H_0 , $\mathbb{E}e_1^2 - \mathbb{E}e_2^2 = -\mathbb{E}(\hat{y}_1 - \hat{y}_2)^2$: the adjustment recentres the differential ([Appendix](#))
 - ▶ approximately Normal for rolling and recursive schemes; the exact critical values of [Clark and McCracken \(2001\)](#) are non-standard

Case study: Meese and Rogoff on EUR/RON

- Meese and Rogoff (1983): structural exchange-rate models do not beat the random walk out of sample at horizons up to one year
 - ▶ their design: rolling out-of-sample forecasts, RMSE against the random walk; the result has survived forty years
- Our data: monthly EUR/RON (BNR reference rate, end of month), $\Delta s_{t+1} = 100 \Delta \ln S_{t+1}$; recursive estimation, forecasts January 2010 to August 2026 ($P = 200$)
 - ▶ S_t : the rate (lei per euro); Δs_{t+1} : its monthly log change in %, positive when the leu depreciates
 - ▶ nested in the random walk ($\Delta \hat{s} = 0$): drift, AR(p), the UIP regression $\Delta s_{t+1} = a + b(i_t^{RO} - i_t^{EA})/12 + u$ Fama (1984)
 - ▶ i_t^{RO}, i_t^{EA} : 3-month interest rates in % a year (divided by 12: per month); a, b : regression coefficients
 - ▶ no estimated parameters: UIP imposed ($a = 0, b = 1$), momentum (mean of the last three changes)
- UIP: uncovered interest parity, the expected depreciation equals the interest differential

EUR/RON: models against the random walk



- Cumulative $\sum_t (e_{RW,t}^2 - e_{m,t}^2)$ Welch and Goyal (2008): a line below zero means model m has accumulated more squared error than the random walk

Interpreting the EUR/RON tests

Model	RMSE / RW	DM (HLN)	CW	p (CW, one-sided)
drift	1.006	0.27	1.06	0.145
AR(1)	1.055	1.13	-0.15	0.561
AR(4)	1.062	1.29	-0.43	0.666
UIP regression	1.016	0.76	0.10	0.458
UIP imposed	1.033	0.99	—	—
momentum	1.183	2.61	—	—
average	1.026	0.85	—	—

- Every model has a higher RMSE than the random walk (random-walk RMSE 0.86 pp a month); the Meese–Rogoff result holds for the leu
- Clark–West does not reject either: the adjustment is not enough to make AR or UIP useful; only the drift gets close
- Managed float: the BNR smooths EUR/RON, so monthly changes are small and close to unpredictable from these predictors

Rationality: Mincer–Zarnowitz and encompassing (1/2)

- ▣ **Mincer–Zarnowitz** Mincer and Zarnowitz (1969): regress the outcome on the forecast
 - ▶ $y_{t+h} = a + b\hat{y}_{t+h|t} + u_{t+h}$
 - ▶ under squared loss an optimal forecast has $(a, b) = (0, 1)$: no bias, and a one-unit forecast change predicts a one-unit outcome change
 - ▶ Wald test of $(a, b) = (0, 1)$ with HAC errors (u is MA($h - 1$)); $b < 1$: forecasts too volatile; $b > 1$: too timid
 - ▶ extended version: add variables in $\mathcal{F}_t$; a significant coefficient means unused information
- ▣ MZ assumes squared loss: under asymmetric loss a rational forecast is biased (Section 1)

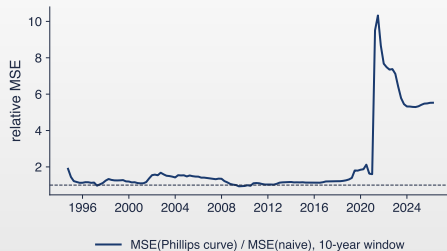
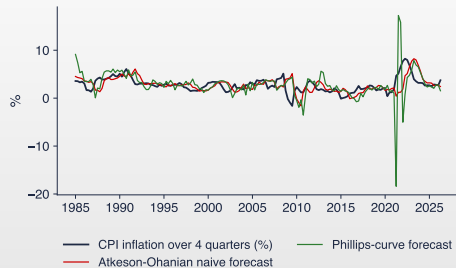
Rationality: Mincer–Zarnowitz and encompassing (2/2)

- ▣ **Encompassing** Chong and Hendry (1986), Harvey, Leybourne and Newbold (1998): forecast 1 encompasses forecast 2 if 2 adds nothing to 1
 - ▶ $y = (1 - \lambda)\hat{y}_1 + \lambda\hat{y}_2 + \varepsilon$, $H_0: \lambda = 0$
 - ▶ λ : the weight of forecast 2 in the best combination; $\lambda = 0$: forecast 2 is useless given forecast 1
 - ▶ HLN test: DM–HLN on $d_t = e_{1t}(e_{1t} - e_{2t})$, one-sided ($\mathbb{E}d_t > 0$ means $\lambda > 0$)
- ▣ If neither encompasses the other, a combination beats both (Section 5)

Case study: Atkeson and Ohanian (2001)

- Atkeson and Ohanian (2001): a naive forecast, “inflation over the next four quarters equals inflation over the last four quarters”, beats Phillips-curve forecasts in the US after 1984
 - ▶ the Phillips curve of Stock and Watson (1999): $\pi_{t+4}^4 - \pi_t^4 = a + \beta(L)u_t + \gamma(L)\Delta\pi_t + e_{t+4}$
 - ▶ π_t^4 : inflation over the last four quarters; π_t : quarterly inflation; u_t : the unemployment rate; $\beta(L), \gamma(L)$: lag polynomials (four lags each)
- Our update: CPI (quarterly average) and the unemployment rate (FRED); four lags of each, rolling window of 15 years; targets 1985Q1 to 2026Q2 ($P = 166$)
 - ▶ October 2025 CPI was not published; the 2025Q4 average uses two months
- Tests: DM–HLN with $h = 4$; GW with $h_t = (1, d_t)$ and $h_t = (1, u_t)$; encompassing in both directions

The naive forecast against the Phillips curve



- Left: four-quarter CPI inflation and the two forecasts made four quarters earlier; right: MSE ratio on a moving 10-year window

Interpreting the Atkeson–Ohanian update

- RMSE: naive 1.69, Phillips curve 2.97 (ratio 1.75); by period: 1.29 (1985–2007), 1.10 (2008–2019), 2.38 (2020–2026)
- Yet DM–HLN = 1.16, p 0.248: the large post-2020 errors inflate the HAC variance as much as the mean
 - ▶ the moving ratio peaks at 10.3 in 2021Q3: the Phillips curve extrapolated the pandemic unemployment spike
- GW: p -value 0.355 with the lagged differential, 0.322 with unemployment
 - ▶ encompassing, H_0 “the naive forecast encompasses the Phillips curve”: p -value 0.289
 - ▶ encompassing, H_0 “the Phillips curve encompasses the naive forecast”: p -value 0.118
- The AO conclusion holds in point estimates, and a test on 41 years of overlapping data cannot separate the two

Many models: data snooping, the reality check and SPA

- Testing k models against a benchmark and reporting the best p -value is **data snooping**
 - ▶ H_0 : $\max_{j \leq k} \mathbb{E}[d_{j,t}] \leq 0$ with $d_{j,t} = L_{0,t} - L_{j,t}$: no model beats the benchmark
 - ▶ $L_{0,t}$: the loss of the benchmark; $L_{j,t}$: the loss of model j ; $d_{j,t} > 0$: model j did better at t
- **Reality check** White (2000): statistic $\max_j \sqrt{P} \bar{d}_j$, p -value from the stationary bootstrap (Chapter 0) under the least favourable configuration
 - ▶ conservative: very poor models inflate the null distribution
- **SPA** Hansen (2005): studentise each $\bar{d}_j$ and drop clearly inferior models from the null distribution (consistent p -value)
 - ▶ lower and upper p -values bracket it; the upper one equals the RC with studentisation
 - ▶ StepM Romano and Wolf (2005) identifies *which* models beat the benchmark
- EUR/RON, $k = 9$ models against the random walk: SPA $p = 0.53$ (lower), 0.85 (consistent), 0.88 (upper): no model beats it

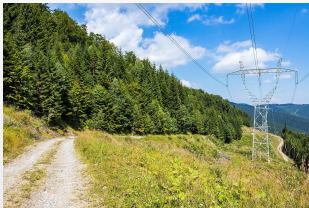
The model confidence set (1/2)

- ▣ **MCS Hansen, Lunde and Nason (2011)**: the set $\hat{\mathcal{M}}_{1-\alpha}$ that contains the best model(s) with probability at least $1 - \alpha$ asymptotically
 - ▶ no benchmark: all models are treated symmetrically; analogous to a confidence interval for a parameter
- ▣ Equivalence test on the m models still in the set
 - ▶ $H_0: \mathbb{E}[d_{i\cdot,t}] = 0$ for all i , $d_{i\cdot,t} = L_{i,t} - \frac{1}{m} \sum_j L_{j,t}$
 - ▶ $d_{i\cdot,t}$: the loss of model i minus the average loss of all models; positive: model i is worse than average
 - ▶ statistic $T_{\max} = \max_i t_i$, $t_i = \bar{d}_{i\cdot} / \widehat{\text{se}}(\bar{d}_{i\cdot})$: the worst model relative to the average, standardised

The model confidence set (2/2)

- Algorithm: start with all models
 - ▶ if the equivalence test rejects, **eliminate** the model with the largest t_i and repeat; stop at the first non-rejection
 - ▶ a block bootstrap gives the null distribution and the standard errors
 - ▶ MCS p -value of a model: the largest test p -value up to its elimination; it stays in $\hat{\mathcal{M}}_{1-\alpha}$ if this is above α
- A large MCS is a finding too
 - ▶ the data cannot separate the models (a weak sample, not equal models)

Case study: day-ahead electricity load in Romania

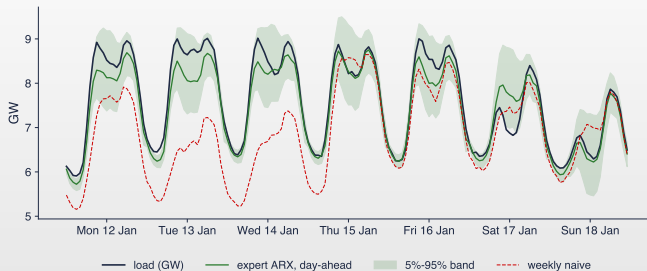


A 400 kV line, Predeluș Pass (2018)

Photo: Drprv (2018); CC BY-SA 3.0; Wikimedia Commons

- Target: the 24 hourly loads of day $d + 1$ with data up to day d ; 1 January 2025 to 30 September 2026 (638 days); mean 5.98 GW
 - ▶ quantile forecasts at the 99 percentiles, scored by the average pinball loss as in GEFCom2014 [Hong et al. \(2016\)](#), [Hong and Fan \(2016\)](#)
- Models: naive day, weekly naive, mean of 4 weeks; the **expert ARX** of [Ziel and Weron \(2018\)](#) for each hour
 - ▶ regressors: $y_{d,h}$, $y_{d-1,h}$, $y_{d-6,h}$, min and max of day d , last hour of d , Monday, Saturday, Sunday; one-year rolling window
 - ▶ quantiles: point forecast plus the empirical quantiles of the last 182 errors at the same hour
- ARX: autoregression with exogenous variables (here calendar dummies)

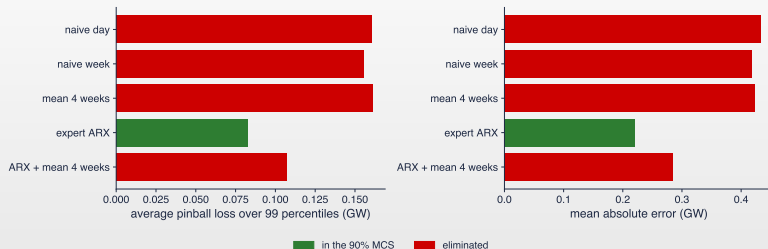
A winter week of load forecasts



- 12–18 January 2026; the weekly naive forecast copies the holiday week of 5–9 January (Orthodox Epiphany and St John, 6–7 January)

ATS_ch1_load_mcs

Average pinball loss and the 90% MCS



□ Daily average losses; MCS with $T_{\max}$, moving-block bootstrap (blocks of 7 days), 2000 replications; green: in the MCS

Q ATS_ch1_load_mcs

Interpreting the load comparison

Model	Pinball (MW)	MAE (MW)	90% coverage	MCS p
naive (day d)	161	433	89.2%	< 0.001
weekly naive	155	417	87.6%	< 0.001
mean of 4 weeks	161	423	88.7%	< 0.001
expert ARX	83	221	88.2%	1.000
ARX + mean of 4 weeks	107	285	88.3%	< 0.001

- The expert ARX halves the losses of the naive rules and is alone in the MCS; DM–HLN against the ARX + mean combination: -10.7
- Here combining with a much worse forecast hurts: equal weights help only when the forecasts are of similar quality (Section 5)
- All intervals under-cover slightly; the largest losses fall on and right after public holidays (8 January 2026, 25 December 2025, 1 January 2025): calendar effects are the next improvement

Recap: Comparing forecasts

- DM compares forecasts; with estimated models choose West (population) or GW (methods, rolling window)
- Multi-step, small P : HAC variance plus the HLN correction and t_{P-1}
- Nested models: Clark–West; many models: SPA or the MCS, never the best p -value
- Romanian inflation, US inflation, EUR/RON: the simple benchmark is hard to beat and differences are rarely significant

Bates and Granger (1969)



Clive W. J. Granger (1934–2009), Nobel
Memorial Prize 2003

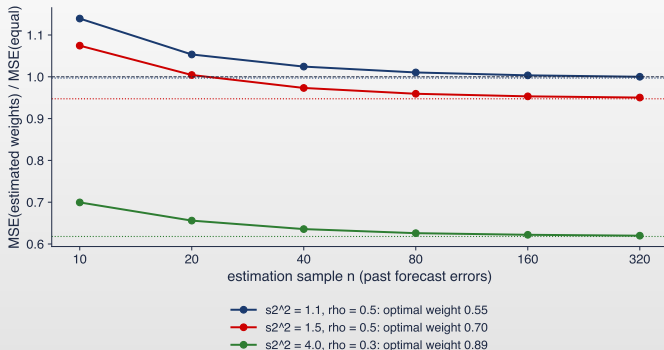
Photo: Olaf Storbeck (2008); CC BY-SA 2.0;
Wikimedia Commons

- Two unbiased forecasts with error variances σ_1^2, σ_2^2 and covariance σ_{12} ; combination $w\hat{y}_1 + (1 - w)\hat{y}_2$ [Bates and Granger \(1969\)](#)
 - ▶ error variance $w^2\sigma_1^2 + (1 - w)^2\sigma_2^2 + 2w(1 - w)\sigma_{12}$, minimised at
 - ▶ $w^* = \frac{\sigma_2^2 - \sigma_{12}}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}}, \quad \sigma_c^2 = \frac{\sigma_1^2\sigma_2^2 - \sigma_{12}^2}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}} \leq \min(\sigma_1^2, \sigma_2^2)$
 - ▶ σ_c^2 : the error variance of the optimal combination, never above that of the better forecast; the less precise forecast gets the smaller weight
- N forecasts: $w^* = \Sigma^{-1}\iota/(\iota'\Sigma^{-1}\iota)$, with Σ the covariance matrix of the errors and ι a vector of ones; [Granger and Ramanathan \(1984\)](#): regress y on the forecasts, with or without a constant and the sum-to-one restriction
 - ▶ combination diversifies model risk, structural breaks and the private information of forecasters [Timmermann \(2006\)](#)

The forecast combination puzzle

- Empirical regularity: the simple average beats the “optimal” estimated weights [Stock and Watson \(2004\)](#), [Timmermann \(2006\)](#), [Genre et al. \(2013\)](#)
 - ▶ Stock–Watson: across 7 countries and many predictors of output growth, equal weights and the median are among the best combinations
- **Explanation** [Smith and Wallis \(2009\)](#), [Claeskens et al. \(2016\)](#): the weights are estimated
 - ▶ $\mathbb{E}[\text{MSE}(\hat{w})] = \text{MSE}(w^*) + \mathbb{E}(\hat{w} - w^*)^2 D$, $D = \sigma_1^2 + \sigma_2^2 - 2\sigma_{12}$ (the variance of $e_1 - e_2$)
 - ▶ $\hat{w}$: the weight estimated from n past errors; MSE: the mean squared error of the combination
 - ▶ gain of w^* over $1/2$: $(w^* - 1/2)^2 D$; loss from estimating: $\text{Var}(\hat{w})D$, of order $1/n$
 - ▶ equal weights win when $(w^* - 1/2)^2 < \text{Var}(\hat{w})$: similar forecasts, short samples, correlated errors
- Also: weights move with breaks, so long windows are biased and short ones noisy

The puzzle by simulation



- Bivariate Normal errors with $\sigma_1^2 = 1$; weights estimated from n past errors; 4 000 replications; dotted: population ratio $\text{MSE}(w^*)/\text{MSE}(1/2)$

Interpreting the puzzle simulation

- Similar forecasts ($w^* = 0.55$): the optimal weight lowers the population MSE only to 0.997 of the equal-weight MSE, while estimating it costs: ratio 1.14 at $n = 10$, 1.024 at $n = 40$
- Moderately different ($w^* = 0.70$): estimated weights need about 20 past errors to break even (1.07 at $n = 10$, 0.973 at $n = 40$)
- Very different ($w^* = 0.89$): estimate the weights, or drop the weak forecast; the load case of Section 4 is of this kind

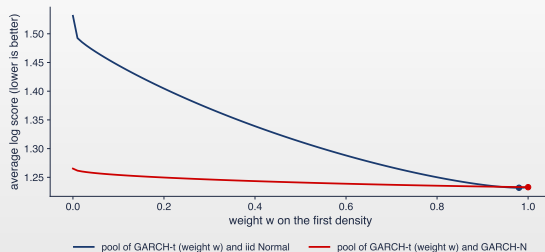
Between equal and optimal: shrinkage and robust combinations

- **Shrinkage:** $\hat{w}_\lambda = \lambda \hat{w} + (1 - \lambda)\iota/N$ Diebold and Pauly (1990), Stock and Watson (2004)
 - ▶ $\hat{w}$: the estimated weights; ι/N : equal weights for N forecasts; $\lambda \in [0, 1]$: $\lambda = 0$ gives the mean, $\lambda = 1$ the estimated weights
 - ▶ a Bayesian prior centred on equal weights; λ falls with the number of forecasts and rises with the sample
- **Performance weights:** $w_i \propto 1/\text{MSE}_i$ on a recent window, ignoring correlations; discounting gives recent errors more weight
 - ▶ **previous best:** select the forecaster with the lowest past MSE (a weight of one)
- **Robust:** median, trimmed mean: protect against outlying forecasters
- Panels of experts change: forecasters enter and exit, so Σ is never fully observed Capistrán and Timmermann (2009)
 - ▶ performance weights need a minimum record; a review of 50 years of practice: Wang et al. (2023)

Combining density forecasts

- **Linear pool:** $p(y) = \sum_i w_i p_i(y)$, $w_i \geq 0$, $\sum w_i = 1$ [Hall and Mitchell \(2007\)](#)
 - ▶ p_i : the density forecast of model i ; w_i : its weight; the pool is a mixture of the densities
 - ▶ weights that minimise the average log score: **optimal prediction pools** [Geweke and Amisano \(2011\)](#)
 - ▶ interior weights even when every model is misspecified: a pool is not a model average in the Bayesian sense
- The linear pool of calibrated densities is **over-dispersed** [Gneiting and Ranjan \(2013\)](#)
 - ▶ the mixture adds the variance of the means; remedy: a beta-transformed pool or quantile averaging (Vincentisation)
- Proper scores (Section 3) are the natural criteria for estimating and comparing pools

Optimal pools for the S&P 500 densities



- Average log score of $w p_{\text{GARCH-}t} + (1 - w) p_2$ on 2013–2026; dots: minima
- With the i.i.d. Normal: $w^* = 0.98$, score 1.2318 against 1.2328 for GARCH- t alone; with GARCH-N: $w^* = 1.00$: GARCH- t dominates, the pool adds almost nothing

Interpreting the optimal pools

- The pool with the i.i.d. Normal puts almost all weight on GARCH- t : a small Normal component only insures against extreme days
- With GARCH-N the optimum is the corner $w = 1$: the two GARCH densities carry the same information, the t tails are simply better
- Weights estimated on the evaluation sample are optimistic; in practice estimate them on a past window and evaluate on the next

Case study: combination schemes against the simple average

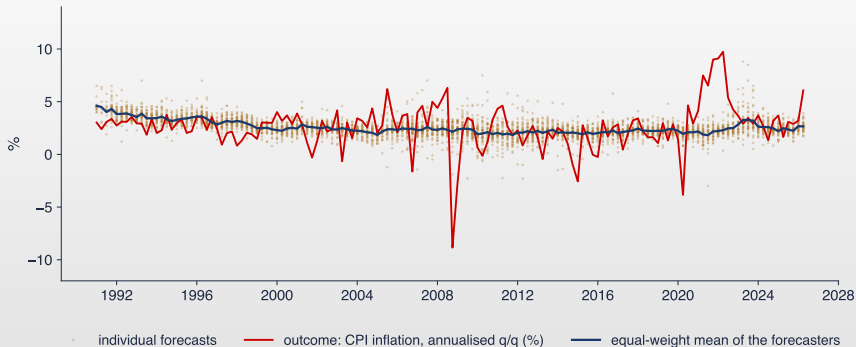


Federal Reserve Bank of Philadelphia, home of the SPF since 1990

Photo: [Beyond My Ken \(2013\)](#); CC BY-SA 4.0;
[Wikimedia Commons](#)

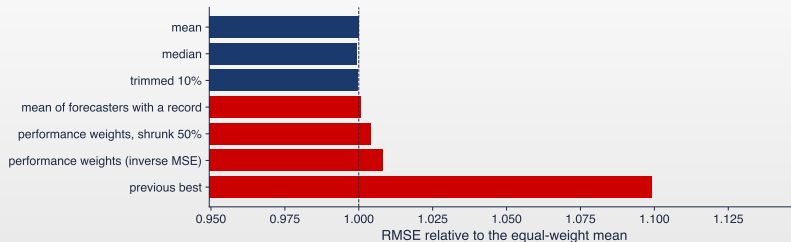
- Genre et al. (2013) compare combination schemes on the ECB Survey of Professional Forecasters: equal weights, median, trimmed mean, performance-based weights, shrinkage, and others
 - ▶ their answer: rarely, and not robustly over time
- We apply their schemes to the US SPF: CPI inflation (annualised q/q) four quarters after the survey quarter; surveys 1990Q1–2025Q2 ($P = 142, 195$ forecasters, on average 36 per survey)
 - ▶ performance weights: inverse MSE over the last 20 surveys with a known outcome, at least 8 past forecasts (on average 25 eligible); trimmed mean: 10% on each side

Individual SPF forecasts and the outcome



- Each dot: one forecaster's CPI forecast for the quarter four quarters ahead, placed at the target quarter; the outcome is far more volatile than any forecast

Combination schemes against the mean



□ RMSE relative to the equal-weight mean (RMSE 2.18 pp); blue: at least as good as the mean

Q.ATS_ch1_spf_combination

Interpreting the SPF combination

- Median 0.999, trimmed mean 1.000, inverse-MSE 1.008, previous best 1.099 (relative RMSE)
 - ▶ DM–HLN against the mean: median p 0.842; inverse MSE p 0.091; previous best HLN 2.76, p 0.007
 - ▶ The answer of [Genre et al. \(2013\)](#) holds on US data: nothing beats the average significantly, and chasing the past winner loses
- Yet 33% of the 76 forecasters with at least 20 forecasts have a lower RMSE than the mean
 - ▶ they are known only ex post, over different and shorter samples: survivorship and luck, not skill one could have selected
- Mincer–Zarnowitz on the mean: $\hat{b} = 0.24$ (SE 0.31), $R^2 = 0.005$, $p < 0.001$: one-year-ahead quarterly CPI inflation is almost unpredictable; the forecasts are smooth, the outcomes noisy

Lessons from the M4 and M5 competitions

- **M4** (2018) [Makridakis et al. \(2020\)](#): 100 000 series, 61 methods, point forecasts and 95% intervals
 - ▶ the winner was a hybrid of exponential smoothing and a recurrent neural network; the second, a combination of statistical methods with weights learned by a meta-learner
 - ▶ combinations dominated the top places; pure machine learning methods did poorly; prediction intervals were too narrow for most methods
- **M5** (2020) [Makridakis et al. \(2022\)](#): 42 840 hierarchical Walmart sales series, weighted scaled errors
 - ▶ gradient-boosted trees (LightGBM) trained *across* series won; exogenous variables (prices, events) and cross-learning mattered
 - ▶ again, ensembles of many models were at the top ([Chapter 12](#))
- Common lesson: combine, evaluate with scale-free scores (MASE, [Chapter 0 of TSA](#)) and proper scores, and report uncertainty that is calibrated

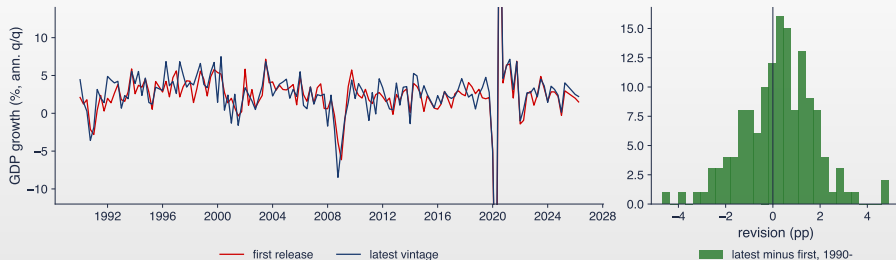
Recap: Combination

- Optimal weights reduce the error variance; estimated weights add a cost of order $1/n$
- The puzzle: when forecasts are similar, equal weights, the median and the trimmed mean win
- Linear pools of densities are over-dispersed; estimate them by the log score
- US SPF: no scheme beats the mean significantly; selecting the past winner loses

Vintages and revisions

- Macroeconomic data are revised: US GDP is first published about 30 days after the quarter and revised for years (benchmark revisions, new methods)
 - ▶ a **vintage** is the data set as available on a given date; the real-time data set stores all vintages [Croushore and Stark \(2001\)](#)
- Sources without a key: ALFRED (St. Louis Fed, vintages of FRED series), the Philadelphia Fed RTDSM; the ECB keeps a real-time database for the euro area; INS publishes a flash estimate of GDP and later revised estimates
 - ▶ forecasts made with latest-vintage data overstate what was predictable in real time [Stark and Croushore \(2002\)](#), [Croushore \(2011\)](#)
- Question for the evaluation: which vintage is the “truth”? The first release (what forecasters target) or the latest (the best estimate of reality)?

US real GDP growth: first release and latest vintage



- Annualised q/q growth, 1990–2026; revisions = latest minus first release (Philadelphia Fed RTDSM); the 2025Q4 first release is missing (government shutdown)

Interpreting the revisions

- Mean revision 0.23 pp, standard deviation 1.54 pp, mean absolute revision 1.21 pp over 143 quarters; correlation first–latest 0.939
 - ▶ largest: 2020Q2 (4.9 pp) and 2008Q4 (−4.7 pp): revisions are large exactly at turning points
- SPF current-quarter nowcast ($P = 143$): RMSE 1.95 against the first release, 2.36 against the latest; MZ slope 1.15 and 1.08
- The ranking of forecasters and models can change with the vintage: state the choice in the pre-registration

A pre-registered evaluation design

- Fix **before** looking at the out-of-sample results
 - ▶ target, horizons, loss or score (Sections 1–3), vintage of the outcome, evaluation period, estimation scheme and window
 - ▶ benchmark (random walk, seasonal naive, survey mean) and the full list of competing models
 - ▶ tests: DM–HLN or GW; Clark–West if nested; SPA or MCS for many models; a correction for several horizons (Holm)
- Report everything that was tried; a model selected on the test period has a biased error
 - ▶ the project of this course starts with exactly such a plan (proposal and pre-registration, 5%)

Recap: Real-time data

- Evaluate forecasts with the data that existed when they were made
- Revisions are large at turning points; the choice of the “truth” vintage changes RMSE and rankings
- Pre-register targets, losses, benchmarks, tests and the vintage

An open question

- Does the forecast combination puzzle survive when the panel of experts changes over time and the outcome distribution shifts (2020–2023)?
 - ▶ formal: $H_0: \mathbb{E}[(y - \hat{y}_\lambda)^2 - (y - \bar{y})^2] \geq 0$ for every shrinkage factor λ fixed in advance, in every sub-period
 - ▶ $\hat{y}_\lambda$: the shrinkage combination with factor λ ; $\bar{y}$: the mean of all forecasters
 - ▶ falsified by a pre-registered λ that beats the mean with DM–HLN $p < 0.05$ after a Holm correction across sub-periods
- Why it matters: central banks publish the survey mean; a better combination would change the published expectations
 - ▶ literature to start from: Genre et al. (2013), Capistrán and Timmermann (2009), Smith and Wallis (2009), Claeskens et al. (2016), Wang et al. (2023)

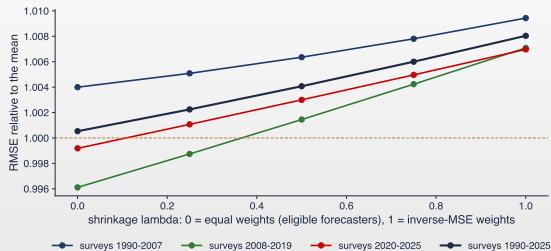
The discovery loop with an AI assistant

- An AI assistant (an LLM such as Claude, ChatGPT, Gemini or Copilot) speeds up each step; Semantic Scholar and Elicit help with the literature
 - ▶ **literature:** List peer-reviewed papers on combining SPF forecasts with entry and exit of forecasters; give DOIs. Then check every DOI on Crossref
 - ▶ **hypothesis:** Propose three reasons why inverse-MSE weights could beat the mean after 2020, each with a test.
 - ▶ **code and replication:** ask for a function, then reproduce a known number first (the relative RMSE of the median in this lecture)
 - ▶ **robustness and critique:** Act as a hostile referee: list every way this evaluation could be data-snooped.
- Report: what was asked, what was kept, what was rejected (AI_USE.md, AI_ERRORS.md)

Required checks

- Every reference exists and says what is claimed (DOI resolves, title matches, the result is in the paper)
- No look-ahead: weights at survey s use only outcomes known at s (here, forecasts from at least five surveys earlier, whose target quarter has ended)
- The HAC lag matches the overlap of the targets ($h - 1$); the HLN correction is applied
- λ , sub-periods and the minimum record are fixed before the results; all variants are reported
- The interpretation separates “not significant” from “equal”

Mini-case: shrinkage across sub-periods



- RMSE of $\lambda \cdot (\text{inverse-MSE weights}) + (1 - \lambda) \cdot (\text{equal weights among eligible forecasters})$, relative to the mean of all forecasters; US SPF, CPI four quarters ahead
- $\lambda = 1$: 1.009 (1990–2007, $P = 72$), 1.007 (2008–2019), 1.007 (2020–2025); full sample HLN 1.70, p 0.091
- Every sub-period: the more weight on past performance, the worse; the puzzle survives the 2020–2023 inflation surge

Project idea

- ▣ **Combining Romanian inflation forecasts:** models (AR, Phillips curve with the output gap, BNR target), surveys and market-based expectations
 - ▶ pre-register horizons 1–12 months, the pinball loss at 5%, 50%, 95% and the CRPS, the Holm correction across horizons
 - ▶ replicate first: the Romanian inflation table of this lecture (Seminar 1, C1 does it by horizon)
 - ▶ extension: density combination by optimal pools; GW test of whether the 2022 shock changed the winner
- ▣ Deliverables follow the course rules: repository, report, AI_USE.md, AI_ERRORS.md, oral defence

Key takeaways

- ▣ Choose the loss or score first: it defines what a “good” forecast is (mean, median, quantile, distribution)
- ▣ Density forecasts: check calibration with PIT, rank with proper scores; ES needs VaR to be scored
- ▣ Tests: DM–HLN with HAC, GW for methods, Clark–West for nested models, SPA and MCS for many
- ▣ Combine similar forecasts with equal weights; estimated weights pay off only with very different forecasts and long records
- ▣ Evaluate in real time and pre-register the design

Self-assessment

Questions

- ▣ Which functional does the pinball loss with $\tau = 0.05$ elicit?
- ▣ What does a U-shaped PIT histogram say about the intervals?
- ▣ Why is DM not valid for nested models with estimated parameters?
- ▣ When do estimated combination weights beat equal weights?
- ▣ Why can ES not be ranked on its own?

Next:

- ▣ Structural breaks and nonlinear models: tests for breaks, threshold models
- ▣ Forecast evaluation under instability: rolling windows, forecast breakdowns, the fluctuation test [Giacomini and Rossi \(2010\)](#)

Appendix: consistency of the pinball loss

- $\mathbb{E}\rho_\tau(Y - x) = \tau \int_x^\infty (y - x)dF(y) + (1 - \tau) \int_{-\infty}^x (x - y)dF(y)$
- Derivative (Leibniz): $-\tau(1 - F(x)) + (1 - \tau)F(x) = F(x) - \tau$; zero at $x = F^{-1}(\tau)$; second derivative $f(x) \geq 0$
- For $\tau = 1/2$: $\rho_{1/2}(e) = |e|/2$, so the median; for x off the quantile the excess loss is $\int_q^x (F(z) - \tau)dz > 0$
- Generalised piecewise linear losses $(1\{y \leq x\} - \tau)(g(x) - g(y))$, g increasing, are the whole class of consistent losses for the quantile [Gneiting \(2011\)](#)

Appendix: the kernel form of the CRPS

- Write $(F(z) - 1\{y \leq z\})^2 = F(z)^2 - 2F(z)1\{y \leq z\} + 1\{y \leq z\}$
- With $X, X' \sim F$ i.i.d.: $F(z)^2 = P(X \leq z, X' \leq z)$ and
 $\mathbb{E}|X - y| = \int [F(z)(1 - 1\{y \leq z\}) + (1 - F(z))1\{y \leq z\}] dz$
- $\frac{1}{2}\mathbb{E}|X - X'| = \int F(z)(1 - F(z)) dz$; subtracting gives $\int (F(z) - 1\{y \leq z\})^2 dz$
- Propriety: $\mathbb{E}_G \text{CRPS}(F, Y) - \mathbb{E}_G \text{CRPS}(G, Y) = \int (F(z) - G(z))^2 dz \geq 0$, zero only if $F = G$

Appendix: the Clark–West adjustment

- Under H_0 the larger model's extra coefficients $\beta_2 = 0$; its forecast is $\hat{y}_2 = \hat{y}_1 + \hat{\beta}_2' x_2$ approximately, with $\hat{\beta}_2$ pure estimation noise
- $e_2 = e_1 - (\hat{y}_2 - \hat{y}_1)$, so $e_1^2 - e_2^2 = 2e_1(\hat{y}_2 - \hat{y}_1) - (\hat{y}_2 - \hat{y}_1)^2$
- $\mathbb{E}[e_1(\hat{y}_2 - \hat{y}_1)] = 0$ under H_0 (e_1 is a martingale difference), hence $\mathbb{E}(e_1^2 - e_2^2) = -\mathbb{E}(\hat{y}_2 - \hat{y}_1)^2 < 0$
- Adding $(\hat{y}_1 - \hat{y}_2)^2$ recentres the differential at zero: $f_t = 2e_{1t}(\hat{y}_{2t} - \hat{y}_{1t})$, the encompassing statistic of [Harvey, Leybourne and Newbold \(1998\)](#) in disguise

References (1/7)

- ▣ Amisano, G., & Giacomini, R. (2007). Comparing density forecasts via weighted likelihood ratio tests. *Journal of Business & Economic Statistics*, 25(2), 177–190. doi:10.1198/073500106000000332
- ▣ Atkeson, A., & Ohanian, L. E. (2001). Are Phillips curves useful for forecasting inflation? *Federal Reserve Bank of Minneapolis Quarterly Review*, 25(1), 2–11. doi:10.21034/qv.2511
- ▣ Banerjee, A., Guo, X., & Wang, H. (2005). On the optimality of conditional expectation as a Bregman predictor. *IEEE Transactions on Information Theory*, 51(7), 2664–2669. doi:10.1109/TIT.2005.850145
- ▣ Bates, J. M., & Granger, C. W. J. (1969). The combination of forecasts. *Operational Research Quarterly*, 20(4), 451–468. doi:10.1057/jors.1969.103
- ▣ Berkowitz, J. (2001). Testing density forecasts, with applications to risk management. *Journal of Business & Economic Statistics*, 19(4), 465–474. doi:10.1198/07350010152596718
- ▣ Bollerslev, T. (1987). A conditionally heteroskedastic time series model for speculative prices and rates of return. *The Review of Economics and Statistics*, 69(3), 542–547. doi:10.2307/1925546
- ▣ Brier, G. W. (1950). Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1), 1–3. doi:10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2
- ▣ Capistrán, C., & Timmermann, A. (2009). Forecast combination with entry and exit of experts. *Journal of Business & Economic Statistics*, 27(4), 428–440. doi:10.1198/jbes.2009.07211
- ▣ Chong, Y. Y., & Hendry, D. F. (1986). Econometric evaluation of linear macro-economic models. *The Review of Economic Studies*, 53(4), 671–690. doi:10.2307/2297611
- ▣ Christoffersen, P. F., & Diebold, F. X. (1997). Optimal prediction under asymmetric loss. *Econometric Theory*, 13(6), 808–817. doi:10.1017/S0266466600006277
- ▣ Christoffersen, P. F. (1998). Evaluating interval forecasts. *International Economic Review*, 39(4), 841–862. doi:10.2307/2527341

References (2/7)

- Claeskens, G., Magnus, J. R., Vasnev, A. L., & Wang, W. (2016). The forecast combination puzzle: A simple theoretical explanation. *International Journal of Forecasting*, 32(3), 754–762. doi:10.1016/j.ijforecast.2015.12.005
- Clark, T. E., & McCracken, M. W. (2001). Tests of equal forecast accuracy and encompassing for nested models. *Journal of Econometrics*, 105(1), 85–110. doi:10.1016/S0304-4076(01)00071-9
- Clark, T. E., & West, K. D. (2007). Approximately normal tests for equal predictive accuracy in nested models. *Journal of Econometrics*, 138(1), 291–311. doi:10.1016/j.jeconom.2006.05.023
- Croushore, D. (2011). Frontiers of real-time data analysis. *Journal of Economic Literature*, 49(1), 72–100. doi:10.1257/jel.49.1.72
- Croushore, D., & Stark, T. (2001). A real-time data set for macroeconomists. *Journal of Econometrics*, 105(1), 111–130. doi:10.1016/S0304-4076(01)00072-0
- Dawid, A. P. (1984). Statistical theory: The prequential approach. *Journal of the Royal Statistical Society, Series A*, 147(2), 278–292. doi:10.2307/2981683
- Diebold, F. X. (2015). Comparing predictive accuracy, twenty years later: A personal perspective on the use and abuse of Diebold–Mariano tests. *Journal of Business & Economic Statistics*, 33(1), 1. doi:10.1080/07350015.2014.983236
- Diebold, F. X., Gunther, T. A., & Tay, A. S. (1998). Evaluating density forecasts with applications to financial risk management. *International Economic Review*, 39(4), 863–883. doi:10.2307/2527342
- Diebold, F. X., & Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263. doi:10.1080/07350015.1995.10524599
- Diebold, F. X., & Pauly, P. (1990). The use of prior information in forecast combination. *International Journal of Forecasting*, 6(4), 503–508. doi:10.1016/0169-2070(90)90028-A
- Ehm, W., Gneiting, T., Jordan, A., & Krüger, F. (2016). Of quantiles and expectiles: Consistent scoring functions, Choquet representations and forecast rankings. *Journal of the Royal Statistical Society, Series B*, 78(3), 505–562. doi:10.1111/rssb.12154

References (3/7)

- Elliott, G., & Timmermann, A. (2008). Economic forecasting. *Journal of Economic Literature*, 46(1), 3–56. doi:10.1257/jel.46.1.3
- Fama, E. F. (1984). Forward and spot exchange rates. *Journal of Monetary Economics*, 14(3), 319–338. doi:10.1016/0304-3932(84)90046-1
- Fissler, T., & Ziegel, J. F. (2016). Higher order elicibility and Osband's principle. *The Annals of Statistics*, 44(4), 1680–1707. doi:10.1214/16-AOS1439
- Genre, V., Kenny, G., Meyler, A., & Timmermann, A. (2013). Combining expert forecasts: Can anything beat the simple average? *International Journal of Forecasting*, 29(1), 108–121. doi:10.1016/j.ijforecast.2012.06.004
- Geweke, J., & Amisano, G. (2011). Optimal prediction pools. *Journal of Econometrics*, 164(1), 130–141. doi:10.1016/j.jeconom.2011.02.017
- Giacomini, R., & Rossi, B. (2010). Forecast comparisons in unstable environments. *Journal of Applied Econometrics*, 25(4), 595–620. doi:10.1002/jae.1177
- Giacomini, R., & White, H. (2006). Tests of conditional predictive ability. *Econometrica*, 74(6), 1545–1578. doi:10.1111/j.1468-0262.2006.00718.x
- Gneiting, T., Balabdaoui, F., & Raftery, A. E. (2007). Probabilistic forecasts, calibration and sharpness. *Journal of the Royal Statistical Society, Series B*, 69(2), 243–268. doi:10.1111/j.1467-9868.2007.00587.x
- Gneiting, T., & Katzfuss, M. (2014). Probabilistic forecasting. *Annual Review of Statistics and Its Application*, 1, 125–151. doi:10.1146/annurev-statistics-062713-085831
- Gneiting, T. (2011). Making and evaluating point forecasts. *Journal of the American Statistical Association*, 106(494), 746–762. doi:10.1198/jasa.2011.r10138
- Gneiting, T., & Raftery, A. E. (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477), 359–378. doi:10.1198/016214506000001437

References (4/7)

- ▣ Gneiting, T., & Ranjan, R. (2013). Combining predictive distributions. *Electronic Journal of Statistics*, 7, 1747–1782. doi:10.1214/13-EJS823
- ▣ Gneiting, T., & Ranjan, R. (2011). Comparing density forecasts using threshold- and quantile-weighted scoring rules. *Journal of Business & Economic Statistics*, 29(3), 411–422. doi:10.1198/jbes.2010.08110
- ▣ Good, I. J. (1952). Rational decisions. *Journal of the Royal Statistical Society, Series B*, 14(1), 107–114. doi:10.1111/j.2517-6161.1952.tb00104.x
- ▣ Granger, C. W. J., & Ramanathan, R. (1984). Improved methods of combining forecasts. *Journal of Forecasting*, 3(2), 197–204. doi:10.1002/for.3980030207
- ▣ Hall, S. G., & Mitchell, J. (2007). Combining density forecasts. *International Journal of Forecasting*, 23(1), 1–13. doi:10.1016/j.ijforecast.2006.08.001
- ▣ Hamill, T. M. (2001). Interpretation of rank histograms for verifying ensemble forecasts. *Monthly Weather Review*, 129(3), 550–560. doi:10.1175/1520-0493(2001)129<0550:IORHFV>2.0.CO;2
- ▣ Hamilton, J. D. (1994). *Time Series Analysis*. Princeton University Press. doi:10.2307/j.ctv14jx6sm
- ▣ Hansen, P. R. (2005). A test for superior predictive ability. *Journal of Business & Economic Statistics*, 23(4), 365–380. doi:10.1198/073500105000000063
- ▣ Hansen, P. R., Lunde, A., & Nason, J. M. (2011). The model confidence set. *Econometrica*, 79(2), 453–497. doi:10.3982/ECTA5771
- ▣ Harvey, D. I., Leybourne, S. J., & Newbold, P. (1998). Tests for forecast encompassing. *Journal of Business & Economic Statistics*, 16(2), 254–259. doi:10.1080/07350015.1998.10524759
- ▣ Harvey, D., Leybourne, S., & Newbold, P. (1997). Testing the equality of prediction mean squared errors. *International Journal of Forecasting*, 13(2), 281–291. doi:10.1016/S0169-2070(96)00719-4

References (5/7)

- Hong, T., & Fan, S. (2016). Probabilistic electric load forecasting: A tutorial review. *International Journal of Forecasting*, 32(3), 914–938. doi:10.1016/j.ijforecast.2015.11.011
- Hong, T., Pinson, P., Fan, S., Zareipour, H., Troccoli, A., & Hyndman, R. J. (2016). Probabilistic energy forecasting: Global Energy Forecasting Competition 2014 and beyond. *International Journal of Forecasting*, 32(3), 896–913. doi:10.1016/j.ijforecast.2016.02.001
- Huang, C., & Petukhina, A. (2022). *Applied Time Series Analysis and Forecasting with Python*. Springer. doi:10.1007/978-3-031-13584-2
- Hyndman, R. J., & Koehler, A. B. (2006). Another look at measures of forecast accuracy. *International Journal of Forecasting*, 22(4), 679–688. doi:10.1016/j.ijforecast.2006.03.001
- Kilian, L., & Lütkepohl, H. (2017). *Structural Vector Autoregressive Analysis*. Cambridge University Press. doi:10.1017/9781108164818
- Makridakis, S., Spiliotis, E., & Assimakopoulos, V. (2022). M5 accuracy competition: Results, findings, and conclusions. *International Journal of Forecasting*, 38(4), 1346–1364. doi:10.1016/j.ijforecast.2021.11.013
- Makridakis, S., Spiliotis, E., & Assimakopoulos, V. (2020). The M4 Competition: 100,000 time series and 61 forecasting methods. *International Journal of Forecasting*, 36(1), 54–74. doi:10.1016/j.ijforecast.2019.04.014
- Matheson, J. E., & Winkler, R. L. (1976). Scoring rules for continuous probability distributions. *Management Science*, 22(10), 1087–1096. doi:10.1287/mnsc.22.10.1087
- Meese, R. A., & Rogoff, K. (1983). Empirical exchange rate models of the seventies. *Journal of International Economics*, 14(1–2), 3–24. doi:10.1016/0022-1996(83)90017-X
- Mincer, J., & Zarnowitz, V. (1969). The evaluation of economic forecasts. In J. Mincer (Ed.), *Economic Forecasts and Expectations* (pp. 3–46). NBER. nber.org
- Newey, W. K., & West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708. doi:10.2307/1913610

References (6/7)

- ▣ Patton, A. J. (2020). Comparing possibly misspecified forecasts. *Journal of Business & Economic Statistics*, 38(4), 796–809. doi:10.1080/07350015.2019.1585256
- ▣ Patton, A. J. (2011). Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1), 246–256. doi:10.1016/j.jeconom.2010.03.034
- ▣ Petropoulos, F., Apiletti, D., Assimakopoulos, V., et al. (2022). Forecasting: theory and practice. *International Journal of Forecasting*, 38(3), 705–871. doi:10.1016/j.ijforecast.2021.11.001
- ▣ Romano, J. P., & Wolf, M. (2005). Stepwise multiple testing as formalized data snooping. *Econometrica*, 73(4), 1237–1282. doi:10.1111/j.1468-0262.2005.00615.x
- ▣ Rossi, B., & Sekhposyan, T. (2019). Alternative tests for correct specification of conditional predictive densities. *Journal of Econometrics*, 208(2), 638–657. doi:10.1016/j.jeconom.2018.07.008
- ▣ Savage, L. J. (1971). Elicitation of personal probabilities and expectations. *Journal of the American Statistical Association*, 66(336), 783–801. doi:10.1080/01621459.1971.10482346
- ▣ Scheuerer, M., & Hamill, T. M. (2015). Variogram-based proper scoring rules for probabilistic forecasts of multivariate quantities. *Monthly Weather Review*, 143(4), 1321–1334. doi:10.1175/MWR-D-14-00269.1
- ▣ Smith, J., & Wallis, K. F. (2009). A simple explanation of the forecast combination puzzle. *Oxford Bulletin of Economics and Statistics*, 71(3), 331–355. doi:10.1111/j.1468-0084.2008.00541.x
- ▣ Stark, T., & Croushore, D. (2002). Forecasting with a real-time data set for macroeconomists. *Journal of Macroeconomics*, 24(4), 507–531. doi:10.1016/S0164-0704(02)00062-9
- ▣ Stock, J. H., & Watson, M. W. (2004). Combination forecasts of output growth in a seven-country data set. *Journal of Forecasting*, 23(6), 405–430. doi:10.1002/for.928
- ▣ Stock, J. H., & Watson, M. W. (1999). Forecasting inflation. *Journal of Monetary Economics*, 44(2), 293–335. doi:10.1016/S0304-3932(99)00027-6

References (7/7)

- Timmermann, A. (2006). Forecast combinations. In G. Elliott, C. W. J. Granger, & A. Timmermann (Eds.), *Handbook of Economic Forecasting* (Vol. 1, pp. 135–196). Elsevier. doi:[10.1016/S1574-0706\(05\)01004-9](https://doi.org/10.1016/S1574-0706(05)01004-9)
- Wang, X., Hyndman, R. J., Li, F., & Kang, Y. (2023). Forecast combinations: An over 50-year review. *International Journal of Forecasting*, 39(4), 1518–1547. doi:[10.1016/j.ijforecast.2022.11.005](https://doi.org/10.1016/j.ijforecast.2022.11.005)
- Weber, S. (2006). Distribution-invariant risk measures, information, and dynamic consistency. *Mathematical Finance*, 16(2), 419–441. doi:[10.1111/j.1467-9965.2006.00277.x](https://doi.org/10.1111/j.1467-9965.2006.00277.x)
- Welch, I., & Goyal, A. (2008). A comprehensive look at the empirical performance of equity premium prediction. *The Review of Financial Studies*, 21(4), 1455–1508. doi:[10.1093/rfs/hhm014](https://doi.org/10.1093/rfs/hhm014)
- West, K. D. (1996). Asymptotic inference about predictive ability. *Econometrica*, 64(5), 1067–1084. doi:[10.2307/2171956](https://doi.org/10.2307/2171956)
- White, H. (2000). A reality check for data snooping. *Econometrica*, 68(5), 1097–1126. doi:[10.1111/1468-0262.00152](https://doi.org/10.1111/1468-0262.00152)
- Ziel, F., & Weron, R. (2018). Day-ahead electricity price forecasting with high-dimensional structures: Univariate vs. multivariate modeling frameworks. *Energy Economics*, 70, 396–420. doi:[10.1016/j.eneco.2017.12.016](https://doi.org/10.1016/j.eneco.2017.12.016)